

Akash Kumar

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EDUCATION

Thapar Institute of Engineering and Technology

M.E. in Artificial Intelligence

Patiala, Punjab

Aug. 2025 – Present

Lovely Professional University

B.Tech in Computer Science & Engineering - CGPA: 7.25

Phagwara, Punjab

Aug. 2019 – May 2023

Kendriya Vidyalaya Sangathan

Intermediate (CBSE)-Score: 88.6%

Bulandshahr, U.P

Aug. 2018 – May 2019

PROJECTS

AI Document Summarizer

FastAPI, NLP, Transformers

- Built an AI-powered document summarization platform supporting PDF, DOCX, and TXT files.
- Designed REST APIs using FastAPI and implemented transformer models (T5, BART).
- Improved summarization quality using hierarchical and extractive NLP techniques.

Real-Time MERN Chat Application

MongoDB, Express, React, Node.js

- Developed a real-time chat application using Socket.IO for live communication.
- Designed MongoDB schemas and implemented encrypted user data storage.

LSTM-Based Portfolio Optimization

Python, LSTM, Machine Learning

- Built an ML-driven portfolio optimization system using LSTM-based volatility forecasting.
- Integrated predictions with Modern Portfolio Theory for dynamic asset allocation.
- Achieved an 18% improvement in Sharpe Ratio over traditional MPT models.

Heart Sound Classification Using CNNs

Deep Learning, CNN, Signal Processing

- Developed a CNN-based model to classify normal vs abnormal heart sounds.
- Extracted MFCC and spectrogram features from the PhysioNet PCG dataset.
- Achieved high accuracy in detecting pathological heart conditions.

INTERNSHIP EXPERIENCE

Software Engineering Intern

India

EPAM Systems

6 Months

- Worked on backend development using Spring and Spring Boot.
- Built MVC-based applications using Thymeleaf.
- Followed enterprise-level coding standards and collaborative workflows.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL

Backend & Web: FastAPI, React, Node.js, REST APIs

Databases: MongoDB, MySQL

Machine Learning: Scikit-learn, TensorFlow, Keras, NLP, LSTM, CNN

Data Engineering: Data Pipelines, ETL, Apache Airflow

Tools: Git, Postman, MongoDB Compass, Jupyter Notebook, Linux